

MSCIT SEM 1 - MONGODB

Chapter 2 Unit 1 - Learning MongoDB by implementing web Application & Using Mongoddb with relational Databases

(1) Explain Collection and different commands.

(1) createCollection() :-

It is used to create the collection in database.

Syntax:-

db.createCollection(name,options) Ex:-
db.createCollection("mytable")

The second argument is optional in this method.

(2) showCollections() :-

It will display all the collections of current database.

(3) insert() :-

It is used to insert document (record) in the collection. Ex:- db.mycollection.insert({"no" :001})

(4) drop() :-

It is used to drop a collection from the database.

Syntax:-

db.collectionname.drop() Ex:-
db.mycollection.drop()

drop method return either true or false. If the collection dropped successfully then returns true otherwise return false.

(2) Mongoddb Datatype.

- String : This is most commonly used data type to store the data. String in mongodb must be UTF-8 valid.

Shree Matrumandir College Of IT & MGMT

- Integer : This type is used to store a numerical value. Integer can be 32 bit or 64 bit depending upon your server.
- Boolean : This type is used to store a Boolean (true/ false) value.
- Double : This type is used to store floating point values.
- Min/ Max keys : This type is used to compare a value against the lowest and highest BSON elements.
- Arrays : This type is used to store arrays or list or multiple values into one key.
- Timestamp : This can be handy for recording when a document has been modified or added.
- Object : This data type is used for embedded documents.
- Null : This type is used to store a Null value.
- Symbol : This data type is used identically to a string however, it's generally reserved for languages that use a specific symbol type.
- Date : This data type is used to store the current date or time in UNIX time format. You can specify your own date time by creating object of Date and passing day, month, year into it.
- Object ID : This data type is used to store the document's ID.
- Binary data : This data type is used to store binary data.
- Code : This data type is used to store JavaScript code into document.
- Regular expression : This data type is used to store regular expression.

(3) Explain document and different commands.

In MongoDB the document is inserted into collection into any database.

(1) insert() :-

For insert the document this method is used.

Syntax :- db.collectionname.insert(document name) Ex:- db.stud.insert (

```
{  
    Rno: "1",  
    Name: "mayur",
```

Shree Matrumandir College Of IT & MGMT

```
Course:"bca",
Detail:
  {Duration: "6 month"
   Trainer:
    "jigneshsir"},
Batch: "2016"
})
```

(2) find() :-

For display the documents you have to use find

method. Ex:- db.stud.find()

(3) update() :-

To update the document or to modify existing document of a collection, update() is used.

Syntax:- db.collectionname.update(selection critearea,
updatedata) Ex:- db.stud.update({'course':'bca'},
{\$set: {'course':'bscit'}})

1

(4) array() :-

To insert multiple document in a collection you have to pass an array of documents to the insert method.

```
Ex:- var
allcourse=[
  {
    Rno:1
    Name:
    "mayur"
    Course: "bca"
    Detail:
      {Duration: "6 month"
       Trainer:
        "jigneshsir"},
    Batch: "2016"
  }
]
```

Shree Matrumandir College Of IT & MGMT

```
{
    Rno:2
    Name:
    "nikhil"
    Course:
    "bca" Detail:
    {Duration: "6 month"
    Trainer:
    "jigneshsir"},
    Batch: "2016"
}
];
```

(5) Insert an array :-

To insert an array in to document you have to use insert method with array name as a parameter. It is also known as bulk insert.

Ex:- db.stud.insert(allcourse);

(6) delete():-

To delete the document from collection, remove() is used. It include the following parameters.

1) deletion criteria:- the criteria according to documents will be remove.

2) justOne :- it remove only one document when set to true or 1.

Ex:- db.stud.remove({rno: 1})
db.stud.remove ({type: "programming language"}, 1)

If you want to remove all the documents from the collection then pass empty{} as a parameter of remove method.

Ex:- db.stud.remove({})

(7) find().limit() :-

To display all records or to display limited records find() & limit() used.

Shree Matrumandir College Of IT & MGMT

The find() return all the document but if you want to display the single document then you have to pass {} with find().

```
db.stud.find({})
```

The limit() is used for retrieve one or two document and field from the collection. It accept number argument.

Syntax:-

```
db.collectionname.find().limit(number)
```

Example:- db.stud.find().limit(2)

(8) skip() :-

It is used to skip the document from starting of the document as per given number argument.

Syntax:- db.collectionname.find().limit(number).skip(number)

Example :- db.stud.find().limit(1).skip(1)

(9) sort() :-

It is used to sort the documents in the collections.

Hear the sorting order is specified as 1 or -1. Hear 1 is used for ascending order And -1 is used for descending order.

Example:- db.stdu.find().sort({"course" :-1})

(10) pretty() :-

This method is used to display the result with formatting in proper way. Syntax:- db.collectionname.find().pretty()

Example:- db.stud.find().pretty()

<u>Question</u>	<u>Answer</u>
Which method is used to create collection?	createCollection()
Which method is used to insert document in the collection? Give the syntax.	Insert() Insert({"column name":data})
How can you delete or remove the collection from database?	Drop()
Which data type is used to store multiple values?	Array
Which data type is used to store document ID.	Object ID
Which data type is used to record document added and modification time?	Timestamp
Which method is used to display all document?	find()
Which method is used to display limited document?	find().limit()
Which method is used to display document with proper format?	find().pretty()

(4) DIFFERENT RDBMS WHERE CAUSE EQUIVALENTS IN MONGODB

<u>Operation</u>	<u>Syntax</u>	<u>Example</u>	<u>RDBMS Equivalen t</u>
Equality	{<key>:<value>}	db.mycol.find ({"by":"tutorialspoint"}) .pretty()	where by = 'tutorialspoint'
Less Than	{<key>:{\$lt:<value>}}	db.mycol.find ({"likes":{\$lt:50}}).pretty()	where likes < 50
Less Than Equals	{<key>:{\$lte:<value>}} }}	db.mycol.find ({"likes":{\$lte:50}}) .pretty()	where likes <=50
Greater Than	{<key>:{\$gt:<value>}} }	db.mycol.find ({"likes":{\$gt:50}}) .pretty()	where likes > 50
Greater Than Equals	{<key>:{\$gte:<value>}} >}}	db.mycol.find ({"likes":{\$gte:50}}) .pretty()	where likes >=50
Not Equals	{<key>:{\$ne:<value>}} }}	db.mycol.find ({"likes":{\$ne:50}}) .pretty()	where likes !=50

(5) Create database, collection and connection with mongodb in php.

```
<?php

    $m=new MongoClient();
    echo "connection mongo client sucessfully";

    $db=$m->mydb;
    echo "Database mydb is selected";

    $collection=$db->createCollection("my
    col");    echo    "collection    created
    successfully";

?>
```

(6) Test Connection

```
<?php

try
{
    $mongo = new MongoClient(); //create a connection to MongoDB

    $databases = $mongo->listDBs(); //List all
    databases echo '<pre>';
    print_r($databases);
    $mongo->close();
}

catch(MongoConnectionException $e)
{
    die($e->getMessage()); //handle connection error

}
```


<u>Question</u>	<u>Answer</u>
Give the syntax of less than cause in mongodb. Give the example also.	<code>{{(<key>:{\$lt:<value>}}}</code> <code>db.result.find({"percentage":{\$lt:60}}).pretty()</code>
Which command is used for display the message?	<code>echo "message"</code>
Which function is used to create mongodb client in php?	<code>MongoClient()</code>
Which function is used to display all databases of mongodb?	<code>listDBs()</code>
Which method is used to end the connection of mongodb and php?	<code>close()</code>

(7) Insert document in emp collection in hns database via php.

```
<?php
$m=new MongoClient();
echo "connection to database sucessfully";
$db=$m->mydb;
echo "Database mydb is selected";
$collection=$db->mycol;
echo "collection created successfully";

$document=array(
"title"=>"MongoDB",
"description"=>"database",
"likes"=>"100",
"url"=>"http://www.hnshukla.com",
"by"=>"hns",
);

$collection->insert($document);
echo "document inserted successfully";
```

?>

(8) Display document from emp collection from hns database.

```
<?php
try
{
    $connection = new MongoClient();
    $database = $connection->selectDB('hns');
    $collection = $database->selectCollection('emp');
}

catch(MongoConnectionException $e)
{
    die("Failed to connect to database ".$e->getMessage());
}
```

```

$cursor = $collection->find();
?>
<?php
echo "<table border=1>";
echo
"<tr><th>EMPNO</th><th>NAME</th><th>SALARY</th>";
while ($cursor->hasNext()):$data = $cursor->getNext();

    echo "<tr><td>";

    echo $data['empno'];
    echo    "</td><td>";
    echo
    $data['empnm'];
    echo    "</td><td>";
    echo $data['salary'];

    echo "</td></tr>";

endwhile;
echo "</table>";
?>

```

(9) Update document from emp collection from hns database.

```

<?php
    $m=new MongoClient();
echo "connection to database sucessfully";

    $db=$m->hns;
    echo "Database is selected";

$collection=$db->emp;
echo "collection created successfully";

$collection->update(array("empnm"=>"mayur"),
array('$set'=>array("empnm"=>"hemang")));

```

```

        echo "document updated successfully";

        $cursor=$collection->find();
        foreach($cursor          as
        $document)
        {
            echo $document["empnm"]."\n";
        }
    ?>

```

(10) Delete document from emp collection from hns database.

```

<?php

$m= new MongoClient();
echo "connection to database suceessfully";

$db=$m->hns;
echo "database selected";

$collection = $db->emp;
echo "collection selected suceessfully";

$collection-
>remove(array"empnm"=>"hemang"));    echo
"documents deleted suceessfully";

$cursor  =$collection->find();
echo "Updated document";

foreach ($cursor as $document)
{
    echo $document["empnm"] . "\n";
}
?>

```

(11) Insert data into database from textbox. Ans:-

empno :-
empname:
-
empsalary:
-

```
<html>
  <body>
<form
action="MongoConnection.php"method="post">

Enter employee no:- <input type="text" name="no"/><br>
Enter employee name:- <input type="text"
name="nm"/><br> Enter employee salary:- <input
type="text" name="sal"/><br>

<input type="submit" name ="insert" value="Insert"/>

</form>
  </body>

</html>
```

```
-----
<?php
$no=$_POST['no'];
$nm=$_POST['nm'];
$sal=$_POST['sal'];

$m = new MongoClient();
$db = $m->hns;
echo "Connection successfully";

$coll = $db->emp;
echo "Database is found is here";
```

```

$data = array('empno' => $no,'empnm' => $nm,'salary'=> $sal);

$coll->insert($data);
echo "Insert Successfully";

header('Location:/mscit1/displaydata.php');
?>

```

(12) Example of Blog Post.

```

<html>
<head>

    <title>Blog Post Creator</title>
</head>
<body>

    <h1>Blog Post Creator</h1>

    <form action="blogpostconn.php" method="post">
        <h3>Title</h3>
        <p>
            <input type="text" name="title" id="title"/>
        </p>
        <h3>Content</h3>
        <textarea name="content" rows="20"></textarea>
        <p>
            <input type="submit" name="btn_submit" value="Save"/>
        </p>
    </form>

</body>
</html>

```

<?php

```
try {
    $connection = new MongoClient();
    $database = $connection->selectDB('myblogsite');
    $collection = $database->selectCollection('articles');

    $article = array(
        'title' => $_POST['title'],
        'content' =>
            $_POST['content'],
        'saved_at' => new MongoDate() // $date
    );

    $collection->insert($article);
}
catch(MongoConnectionException $e)
{
    die("Failed to connect to database ".$e->getMessage());
}
catch(MongoException $e)
{
    die('Failed to insert data '.$e->getMessage());
}
?>
<html>
<body>
    Article saved. _id:<?php echo $article['_id'];?>.
    <a href="blogpost.php">
        Write another one?</a>
</body>
</html>
```

<u>Question</u>	<u>Answer</u>
How can you create the collection student in mydb database?	<code>\$collection=\$mydb->student;</code>
How can you select the collection student from mydb database?	<code>\$collection=\$mydb->selectCollection('student');</code>
Which method is used to print the message and exit from the current php script.	<code>die()</code> method
Which method is used to delete the document from collection?	<code>remove()</code>
Which method is used to fetch system error message in catch block?	<code>getMessage()</code>

(13) Explain MongoDB and RDBMS together.

Any relational database has typical style and design that shows number of tables and the relationship between these tables.

MongoDB is document database which is based on collection in any database. In MongoDB there is no fixed structure or no any complex join like RDBMS.

MongoDB support big data, mobile and social infrastructure, user data management, content management and data hub.

MongoDB support array as first class object of a document so we can easily handle many to one or many to many relationship.

MongoDB also support embedded document and nested document.

MongoDB also support denormalization of your data.

In RDBMS mysql is somehow similar to MongoDB.

MongoDB and mysql are both capable of representing rich data model.

Where mysql use fixed design table but MongoDB has design free document. Mysql and MongoDB both support B-tree index with similar behavior.

MongoDB and mysql both support replication.

MongoDB also support reference document so one document may also a reference to another document.

(14) Explain how to manage a relationship between documents in MongoDB.

Mongodb relationship shows that how different documents are related to each other logically.

This relationship can be design through referenced relationship and embedded relationship.

This relationship can be one to one, one to many, many to one or many to many. For example:-

- suppose we have blog application then each post has a title that is one to one relationship.
- Each author and blog is one to many relationships.
- And any post has many tags and the tags refer to many post so that is many to many relationship.

Reference Relationship:-

In the reference relationship method, it includes standard relationship both the client and location documents will be kept independent.

The client document contents a field that will reference the location document's id field

Mongodb support the structure data format into collections of JSON document.

The JSON document is useful for data management and include human readable data format.

A JSON support rich and more flexible data structure then tables.

If we use JSON document in our database then we don't need any object relational mapped between our database and the application.

JSON stands for Java Script Object Notation.

It is open standard, lightweight, language-independent data interchange format based on JavaScript. JSON makes it easy to share data between devices like clients and servers.

Embedded Relationship:-

In the embedded method, we will embed the address document inside the client document.

In this method, all the related information keeps in a single document, which makes it simple to recover and keep up-to-date. The entire document can be recovered in a single query.

<u>Question</u>	<u>Answer</u>
Mongodb support embedded document. True or False	True
Mongodb support de normalization of your data and replication. True or False	True
Mongodb and Mysql both database capable or representing rich data model. True or False	True
Mongodb support reference document. True or False	True
Mongodb require fix structure like RDBMS. True or False	False
What is embedded relationship?	All related information keeps in a single document. It embed the address document inside the client document.
Why embedded relationship is require?	It is helpful to recover document with single query and maintain up to date data.

What is Reference relationship?	It keeps independent documents from client. the location reference will be provide to client with document's id.
Give the full form of JSON.	Java Script Object Notation